

## Do Now | Check your starting point

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

**Using precise computer science knowledge to explain answers, solve problems and recognise what we need to practise next.**

Answer alone before checking. These five classroom checks total 10 marks. Keep your original answers. Record any help you use.

### **Question 1 | Embedded systems | 2 marks**

A washing-machine controller is built into the machine to control washing cycles. A laptop runs email, spreadsheets and games. Explain why the controller is an embedded system and state one difference in purpose from the laptop.

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### **Question 2 | Networks | 2 marks**

A media building is 350 m from a server room. The link carries large video files and passes near electrical equipment. Choose copper or fibre and explain one relevant reason for your choice.

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### **Question 3 | Cyber security | 2 marks**

A computer has missed security updates for six months. Explain how installing the updates can reduce the risk of an attack.

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### **Question 4 | Databases and SQL | 2 marks**

Booking has fields BookingID, MemberID and Session. Explain the effect of this command: UPDATE Booking SET Session = "PM" WHERE BookingID = 502.

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### **Question 5 | Core programming | 2 marks**

All inputs are integers. Valid Day values are 1 to 7 inclusive. Write a Boolean condition that is true for an invalid Day and give one invalid value.

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# Embedded systems | Read and connect

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

**WAGBA: Using precise computer science knowledge to explain answers, solve problems and recognise what we need to practise next.**

Specification 3.4.4 | Chapter 7.1: Embedded systems, pp. 202–203

An embedded system is a computer system that forms part of a larger electrical or mechanical system. It is designed to carry out one task or a limited set of specific tasks. A washing-machine controller monitors inputs and controls the washing cycle; it is not intended to run an arbitrary range of applications.

A general-purpose computer, such as a desktop computer, can run different applications for different purposes. Both kinds of system can contain a processor and memory. Having a processor does not distinguish one from the other: their purpose and how they are used do.

Embedded does not mean “simple”, “not connected to the internet” or “unable to receive updates”. Some embedded systems are complex and connected. For an exam comparison, use a clear paired difference: a dedicated control purpose versus running a range of applications.

## Worked example

A controller built into a lift reads button and position-sensor inputs and controls the motor and floor display. A laptop can run a spreadsheet, browser and image editor. Explain their different purposes rather than claiming that only one has memory.

**Explain to your partner: what is the key idea, and which common mistake would it prevent?**

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## Embedded systems | Explain, check, improve

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

A dishwasher has a built-in controller. It reads door and temperature sensors and controls the heater and water pump. A technician also uses a laptop for email, reports and diagnostics. Explain why the dishwasher controller is embedded and give a precise comparison with the laptop.

**What larger system contains the controller? What specific purpose does it serve?**

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**Write one paired difference about the software or range of tasks.**

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**Discuss and check. Write one correction and why it changed.**

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**Would adding Wi-Fi to the dishwasher make its controller general-purpose? Explain.**

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# Networks | Read and connect

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

**WAGBA: Using precise computer science knowledge to explain answers, solve problems and recognise what we need to practise next.**

Specification 3.5 | Chapter 8: Networks and protocols, pp. 226–239; network security, pp. 248–249

A computer network connects computers or other devices so that they can communicate and share resources. A LAN normally covers a small geographical area and is usually managed by one organisation. A WAN connects devices across a wider geographical area; geographical scale matters more than the number of devices.

In a star topology, each device connects to a central switch. If one device cable fails, the other connections can still work. If the central switch fails, communication through it is affected. In a bus topology, devices share a backbone cable. A backbone failure can prevent communication across the network. Be explicit about which component fails.

Copper carries electrical signals and is often economical for shorter fixed connections. Fibre carries light and is suitable for higher-bandwidth or longer links and is resistant to electromagnetic interference. Wireless supports mobile devices but has coverage, interference and security considerations.

A protocol supplies rules for communication. SMTP sends email; IMAP accesses server-held email; HTTPS protects web communication using encryption. In the four-layer TCP/IP model, applications use the application layer, TCP/UDP operate at transport, IP at network, and network hardware at link. Do not confuse a transport-layer protocol with a web protocol.

## Worked example

For fixed office desktops near a cabinet, copper may be economical. For a 350 m connection near electrical equipment carrying large video files, fibre is more suitable. "Faster" alone is less convincing than linking bandwidth, distance and interference to the stated requirements.

**Explain to your partner: what is the key idea, and which common mistake would it prevent?**

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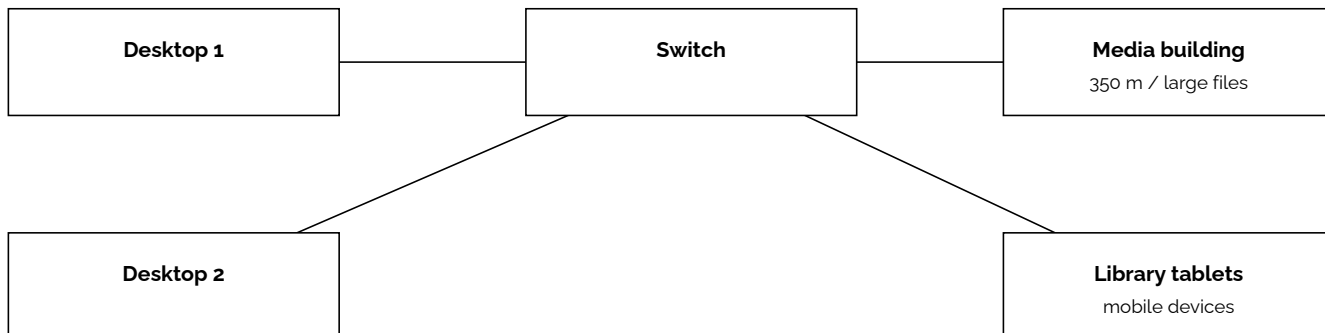
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# Networks | Explain, check, improve

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

A school LAN uses a star topology. Each of 12 desktops has its own cable to a switch. A media building 350 m away transfers large video files and is close to electrical machinery. Mobile tablets are used in the library. Annotate the supplied diagram and justify suitable connections.



Two example desktops shown; the school has 12. Label the media link and tablet connection.

**Explain separately what happens if one desktop cable fails and if the switch fails.**

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**Choose a medium for the media-building link and a method for the tablets. Link each choice to the scenario.**

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**Discuss and check. Write one correction and why it changed.**

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**Two distant campuses each have a LAN. What type of network connects the campuses? What does IMAP do?**

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# Cyber security | Read and connect

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

**WAGBA: Using precise computer science knowledge to explain answers, solve problems and recognise what we need to practise next.**

Specification 3.6–3.6.2 | Chapter 9: Security, pp. 240–253

Cyber security uses processes, practices and technologies to protect networks, computers, programs and data from attack, damage or unauthorised access. Naming a threat is only the beginning; explain how it can cause harm.

Phishing uses deceptive messages to persuade someone to disclose information or follow a harmful link. Pharming redirects users to a fraudulent website. Blagging uses an invented story to obtain information, and shoulder surfing involves observing confidential information. Distinguish how the attack reaches the victim.

Malware is malicious software. A virus attaches to other programs or files and can spread when infected code runs. A worm can spread across networks; a Trojan appears legitimate while hiding harmful functionality. Unpatched software can retain known vulnerabilities, and incorrectly assigned access rights can allow someone to view or change data they should not access.

Choose protection for the mechanism: staff training and checking a message through a trusted route address phishing; updates patch vulnerabilities; appropriate access rights limit unauthorised actions. Authentication checks identity; encryption protects data from being understood without the necessary key. Penetration testing is authorised testing for exploitable weaknesses, not permission for students to attack a live system.

## Worked example

“Use security” is too vague. A stronger explanation is: automatic updates install patches for known vulnerabilities, reducing the opportunity for an attacker to exploit those weaknesses.

**Explain to your partner: what is the key idea, and which common mistake would it prevent?**

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# Cyber security | Explain, check, improve

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

An employee receives an email claiming to be from the school IT team. It links to a look-alike login page and asks for a password. Separately, a shared computer has not received security updates for six months. Identify the two risks and explain an appropriate response to each.

**Explain how the deceptive message could lead to unauthorised access.**

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**Explain why updates address the second risk; distinguish this from protecting against the deceptive email.**

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**Discuss and check. Write one correction and why it changed.**

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**An administrator gives all users permission to delete payroll records. Why is a strong password alone insufficient?**

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# Databases and SQL | Read and connect

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

**WAGBA: Using precise computer science knowledge to explain answers, solve problems and recognise what we need to practise next.**

Specification 3.7.1–3.7.3 | Chapter 10: Database design and SQL, pp. 254–271

A table stores records with the same fields. A primary key uniquely identifies each record. A foreign key is a field used to reference a key in another table; in the simple designs used here it references that table's primary key. Foreign-key values may repeat: many bookings can belong to one member.

Storing a member's contact details once in a Member table avoids copying them into every booking. Repeated data is redundancy. Conflicting copies, such as two different phone numbers for the same member, create inconsistency. A relational design reduces these problems by storing related data in linked tables.

SELECT chooses output fields, FROM identifies source tables, and WHERE selects matching records. A two-table query also needs the linking condition: Member.MemberID = Booking.MemberID. Without it, unrelated records can be combined.

INSERT INTO adds a record; UPDATE changes existing records; DELETE FROM removes records. Use a precise WHERE condition when changing only particular records. UPDATE Booking SET Session = "PM" WHERE BookingID = 502 changes one booking. Without WHERE, every booking would change. Use the exact field and table names given in the question.

## Worked example

SELECT Member.Name, Booking.Session FROM Member, Booking WHERE Member.MemberID = Booking.MemberID AND Booking.Session = "PM" returns member names and sessions for the matching PM bookings, not all possible combinations of members and bookings.

**Explain to your partner: what is the key idea, and which common mistake would it prevent?**

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# Databases and SQL | Explain, check, improve

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

The Member and Booking tables are shown below. Explain their relationship. Write a query listing each member name and session for PM bookings, then write a command changing only BookingID 502 to the AM session. Predict the records affected before checking.

## Member

| MemberID | Name |
|----------|------|
| 11       | Ali  |
| 12       | Bea  |

## Booking

| BookingID | MemberID | Session |
|-----------|----------|---------|
| 501       | 11       | PM      |
| 502       | 12       | PM      |
| 503       | 11       | AM      |

Both commands start from these original records.

**Identify both primary keys and the foreign key. Explain why the foreign key can repeat.**

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**Write the SELECT query and the UPDATE command. Explain the linking condition and WHERE condition.**

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**Discuss and check. Write one correction and why it changed.**

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**Why should Session not be the primary key of Booking? What would happen if the UPDATE had no WHERE?**

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# Core programming | Read and connect

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

**WAGBA: Using precise computer science knowledge to explain answers, solve problems and recognise what we need to practise next.**

Specification 3.1.1; 3.2.2, 3.2.4–3.2.5, 3.2.12 | Chapters 2 and 3: selection, iteration, validation, testing and trace tables

Assignment stores a value in a variable. Selection chooses which instructions execute; iteration repeats instructions. When tracing, follow execution order rather than guessing the final result. Record the initial values and then each change. Read whether a loop includes its end value.

A range check tests whether a value lies within stated limits. If valid integers are 1 to 7 inclusive, the invalid condition is  $\text{Day} < 1 \text{ OR } \text{Day} > 7$ . OR is needed because either violation is invalid. Using AND would require a number to be below 1 and above 7 at the same time.

Normal test data is typical valid input, such as 4. Boundary tests examine the stated limits and values close to them: 1 and 7 should be accepted, while 0 and 8 should be rejected for this range. A value outside the permitted range is also erroneous data. State the expected result for every test; a label alone is not enough.

For a question that states input is an integer, solve the stated numeric validation problem. Do not add string conversion or exception-handling requirements that the question has not asked for. A validation loop must request a new input after an invalid value, not endlessly test the same invalid value.

## Worked example

For valid integer scores 0 to 20 inclusive,  $\text{Score} < 0 \text{ OR } \text{Score} > 20$  identifies invalid values. Test 10 (accept), 0 and 20 (accept), -1 and 21 (reject). Follow the exact range and assumptions in the question.

**Explain to your partner: what is the key idea, and which common mistake would it prevent?**

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# Core programming | Explain, check, improve

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

First trace the supplied algorithm with Values = [2, 7, 4]. Record initial Total and every update. Then plan a flowchart that requests an integer Day from 1 to 7 inclusive, rejects out-of-range values and requests another input until the value is valid. All inputs in this task are integers.

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Values <- [2, 7, 4]
Total <- 0
FOR Index <- 0 TO 2
    IF Values[Index] > 3 THEN
        Total <- Total + Values[Index]
    ENDIF
ENDFOR
OUTPUT Total
  
```

| Index   | Value | Total | Reason |
|---------|-------|-------|--------|
| Initial |       |       |        |
| 0       |       |       |        |
| 1       |       |       |        |
| 2       |       |       |        |

Complete the trace table and explain why only some values increase Total.

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Write the invalid condition, show where input is requested again, and give test data with expected outcomes.

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Discuss and check. Write one correction and why it changed.

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A valid Day has been entered, but the program asks for input forever. What part of the algorithm would you inspect first?

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# Learning Pit Stop | My next practice

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

**WAGBA: Using precise computer science knowledge to explain answers, solve problems and recognise what we need to practise next.**

| Type of learning | What I need                                            |
|------------------|--------------------------------------------------------|
| Knowledge        | Recall the terms and rules.                            |
| Skills           | Perform a trace, query or precise explanation.         |
| Understanding    | Explain why and apply the idea in a different context. |

**My current phase: New learning / Consolidating / Treading water / Drowning**

The specific skill and evidence from my work:

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Previously I thought... I now understand... because...

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Next skill to practise, question/source ID and review date:

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Keep your original attempt and correction separate. Save your evidence for the next review.